RealQM for Molecules: Validation in Progress

Claes Johnson[1]

[1]KTH Royal Institute of Technology, Stockholm, Sweden , claesjohnson@gmail.com

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Mathematical formulation by the author; implementation and numerical experiments developed with AI assistance (Claude, Anthropic).

Abstract

This is a report on work in progress: the validation of RealQM on molecules, a very large class of systems. We present first what works directly, on a selection of basic examples, and indicate briefly the cases that do not yet work, with the result found for each. RealQM [1] describes a molecule as a partition of space into non-overlapping domains, each holding one electron (or a pair counted once) as a real density, with the domain boundaries placed where the density is continuous and each electron free at its wall (a homogeneous Neumann condition). We solve this *hard-partition* model as it stands, converged in the mesh, with nothing adjusted to molecular data (each core radius is set once per atom, from its ionisation energies), using a three-dimensional solver (`molecule.js`) and, for linear molecules whose boundaries are fixed by symmetry or construction (homonuclear X_2, where the boundary is the midplane), an exact axisymmetric reduction that serves as an independent check: on H_2 the two give binding energies of 0.2251 and 0.2241 E_h at the same mesh. Each electron pair can be carried in two parameter-free constructions to choose from: *2split*, two one-electron domains (complete, permanent avoidance), and *2hom*, one pair domain (no avoidance beyond the mean field, the Hartree–Fock level). Both are reported; for H_2 and He the exact value lies between them. Where the boundary is fixed by symmetry the bond is sharp: H_2 -2.3%, converged. A free boundary charges nothing for the valence pressing into a shell, so inner shells are excluded by a fixed core radius per atom: every all-electron LiH with free core boundaries collapses, while the pseudo-Li core with one 2hom bonding pair gives $R_e = 2.85\,a_0$ (exact 3.015), $D_e = 0.095\,E_h$ (exact 0.092) and a dipole of the right sign, 11% high. Within the present selection the model is judged by geometry: bond lengths of hydrides (LiH, BH, BeH_2, OH), of X_2 from Li_2 to N_2 and of linear molecules (HCN, CO, CO_2) are within a few per cent, and the residual forces at the observed geometry are small. Cases that do not yet work—heavy atoms carrying their whole valence (CH_4, NH_3, H_2O), shapes and angles, O_2 and F_2, multiple bonds, dipoles, reaction energies—are listed at the end. Every energy above five domains is checked against an independent recomputation from the converged density. Molecules are built by successive reduction—all electrons, frozen $1s^2$ core, reduced dimension for X_2, reduced valence charges on a coarse mesh—and the last step brings proteins of over two thousand atoms within computational reach, on one GPU in a web browser.

1 The model

RealQM [1] replaces the many-electron wave function by one-electron densities on non-overlapping domains of ordinary three-dimensional space. A molecule with nuclei $\{(\mathbf{R}_a, Z_a)\}$ is described by

domains Ω_k partitioning space, each carrying $n_k \in \{1, 2\}$ electrons (two only of opposite spin) as a real amplitude u_k with $\int_{\Omega_k} u_k^2 = 1$ and density $\rho_k = n_k u_k^2$. The energy is

$$E = \sum_k n_k \Big(\tfrac{1}{2} \int_{\Omega_k} |\nabla u_k|^2 - \int_{\Omega_k} K u_k^2 \Big) + \sum_k \tfrac{n_k(n_k-1)}{2} U_{kk} + \sum_{k<l} n_k n_l U_{kl} + V_{KK}, \tag{1}$$

with $K = \sum_a Z_a / |\mathbf{x} - \mathbf{R}_a|$, U_{kl} the Coulomb interaction of the unit densities u_k^2 and u_l^2, and V_{KK} the nuclear repulsion. A pair domain ($n_k = 2$) counts its single pair once.

Boundary conditions. Each u_k satisfies a homogeneous Neumann condition on $\partial\Omega_k$: the kinetic energy is counted only inside the domain, and the amplitude at the wall is free. The boundaries are placed by *density continuity*, $\rho_k = \rho_l$ across each interface. Energy minimisation over the boundary position is not used: in this model a ragged or displaced boundary can lower the energy (Section 5.3), and only continuity holds it.

2 Atoms

Atoms are computed in spherical symmetry, with all electrons organised in non-overlapping radial shells following the Atom Simulator configurations [2] (Li 2+1, B 2+3, ...): a one-dimensional problem in the radius, with each shell boundary free and placed by the same continuity rule as in three dimensions. For use in molecules the atom is then reduced to a *pseudo-atom*: the $1s^2$ core (and further inner shells) is frozen into a pseudo-nucleus of charge Z_x with a hard core of radius r_c, and only the outer shell is carried in three dimensions, split into domains of at most two electrons (boron: pseudo-B +3 with an outer 2hom pair and one electron). A radial shell holding three or four electrons is a shared density in one dimension, which no domain may be in three dimensions; it stands for the spherical average of its angular split into domains of at most two (boron's 3 as 2_{hom}+1, a 4 as a tetrahedron of single electrons). Shells hold at most four—the pair and the tetrahedron, an octet being 4+4, never a single shell of six or eight. Energy does not select the configuration: more sharing always lowers the one-dimensional energy (Li as one shell of 3 is 13.7% too deep, as 2+1 within 0.5%), so the configuration is set by the Pauli cap of two per domain, not by minimisation. The core radius is a property of the atom, not of any molecule, and it is obtained in two ways. *Fitted*: the radius at which the pseudo-atom alone, in three dimensions, reproduces the sum of the ionisation energies of the electrons it carries. *Computed*: the core–valence interface of the one-dimensional all-electron atom, which for a $1s^2$ core is fixed by the model itself, independent of the box:

	r_c fitted to IE (3D)	r_c computed (1D, boxes 6–20 a_0)	difference
Li (2+1)	1.99	1.98–2.05	within 3%
B (2+3)	1.08	0.92–1.00	8–17%
Be (2+2)	1.37	pending	
C (2+4)	1.0	pending	

For Li the two agree, and the core radius is an output of the model. For B the computed radius is shorter than the fitted one. We use the fitted radius throughout, and state the computed one beside it; an outer core shell beyond $1s^2$ (Na and below) has no box-independent interface in one dimension, and there only the fitted radius exists. At $h = 0.1$ the hard core excludes whole cells, so the fitted radius is defined to a few hundredths of a_0.

3 Molecules by successive reduction

Molecules are built by reducing the full model step by step, each step tested on systems where the step before it can still be run.

1. *All electrons* (H_2, He): every electron in its own domain or in a pair, with no pseudo-nucleus. Here an electron pair has two constructions to choose from, 2split and 2hom (Section 5).

2. *Frozen* $1s^2$ *core*: for every molecule beyond H_2 the atoms enter as pseudo-atoms (Section 2); only valence electrons are computed, in domains of at most two, with the inner boundary r_c fixed and all outer boundaries free. A free boundary charges nothing for crowding, so the core sphere must be fixed (Section 6.1). LiH and BH are computed this way.

3. *Reduced dimension* for linear molecules whose boundaries are fixed planes—the midplane of a homonuclear X_2, by symmetry, or planes through the nuclei, by construction: the axisymmetric solver (Section 4).

4. *Large molecules*: pseudo-atoms with reduced valence charges on a coarse mesh. The same `molecule.js` has been run on a 238-residue protein (green fluorescent protein, 2307 atoms, 300^3 grid, $h = 0.4\,a_0$) and on crambin (540 atoms, 200^3, $h = 0.25\,a_0$) on the GPU of a laptop in a web browser. This shows that large molecules are within computational reach; it is not a validation. At this mesh the bond is not resolved, the folds shown use contact restraints, and at this size the solver uses the total-Poisson path for the electron repulsion (Section 4), which must be repaired before results there can be judged.

4 Solvers, and the check between them

`molecule.js`. One amplitude field on a Cartesian grid, each cell labelled by its owning domain, so the domains are disjoint by construction. The kinetic stencil omits every face joining two labels (true Neumann). Boundaries move by label flips: a boundary cell changes owner when the neighbouring domain's density exceeds its own. The Coulomb potential of the other electrons is a direct per-electron Poisson solve. The nuclear kernel is clamped, $-Z/\max(r, R_{\text{sing}})$, with $R_{\text{sing}} = 2h$ unless pinned. The amplitude is relaxed in imaginary time; convergence is declared when the mean energy over a window of 3000 steps changes by less than $10^{-4}\,E_h$ between successive windows, since a free boundary keeps flipping single cells indefinitely and a step-to-step test either never fires or fires early.

Axisymmetric reduction. For a homonuclear X_2 the domain boundary is the midplane by symmetry, and the model reduces exactly to (z, ρ) with a plane boundary. The same holds for any linear molecule whose walls are all fixed planes, such as planes through the nuclei separating the halves of a pseudo-atom's valence, with every bond a 2hom pair (Section 6.3). We use it only there: a free wall between unlike atoms curves, and a plane would impose it. The Neumann condition is imposed by assembling the stiffness only over faces inside a domain. Each domain's lowest state is an exact sparse eigensolve, and the potentials are iterated to self-consistency. Mesh ladders down to $h = 0.05$ are cheap.

The check. The electron–electron energy of the two solvers agrees to 1.4%, and the total energies to the extent their kernels allow. At $R = 2.0\,a_0$, $h = 0.1$ and box $10\,a_0$:

	$E(\mathrm{H_2})$	$2E(\mathrm{H}) - E(\mathrm{H_2})$
`molecule.js` (3D, free boundary)	−1.20734	0.2251
axisymmetric (2D, exact eigensolve)	−1.23685	0.2241
exact	−1.1381	0.1381

The totals differ by the kernel (clamped at $2h$ in 3D, analytic in 2D), which largely cancels in the binding energy; the binding energies agree to 0.5%. The three-dimensional solver reproduces the model.

An independent energy check. The agreement above is for two domains. With more than five, `molecule.js` by default replaces the per-electron Poisson solves by one total Poisson solve with a self-interaction correction, and that path had not been checked. We therefore recompute every energy from the converged state alone: the amplitude and the domain label of each cell are exported, and V_{ee} is recomputed by a zero-padded FFT convolution with $1/r$ (each pair of domains once, and $(n-1)/n$ of a pair domain's own Hartree energy), V_{eK} from an analytic kernel, T from finite differences inside each domain with free walls, and V_{KK} as a point sum. On the H atom this agrees with the solver to 10^{-5}, on H_2 to $0.008\,E_h$. On CH_4 it exposed the total-Poisson path: for its own converged density the true V_{ee} is 10.28 where the path reports 7.81, so it reported an energy $2.5\,E_h$ too low and also converged to a worse density (true energy −9.00 against −9.27 on the per-electron path). All results here use the per-electron path. That path over-counts V_{ee} slightly, by $0.04\,E_h$ for the C atom to 0.36 for ethane, growing with the number of pairs; every energy quoted for more than two domains is the independently recomputed one.

5 Hydrogen and helium: two constructions of a pair

5.1 Two constructions of an electron pair

RealQM carries no spin variable; spin enters only through the Pauli principle, as the rule that a domain holds at most two electrons, and two only if of opposite spin. Electrons in different domains are excluded from each other by disjointness, whatever their spin. A pair of opposite spin can therefore be carried in two parameter-free ways. In the *2split* construction each electron owns its own domain and the two meet at a free wall: their avoidance is complete and permanent. In the *2hom* construction both electrons share one domain and one density, the overlay that Pauli permits for opposite spins, each feeling half its Hartree field: there is no avoidance beyond the mean field, and the pair is a restricted Hartree–Fock pair. A one-dimensional radial check on Li^+ makes the last point exact: the 2hom $1s^2$ pair gives −7.23630 against the Hartree–Fock −7.2364. The true state avoids partially and dynamically.

5.2 H_2 and He

For both two-electron systems the exact value lies between the two constructions (H_2 axisymmetric, $R = 1.4$, $h = 0.1$):

H_2	T	V_{eK}	V_{ee}	D_e
2split, two one-electron domains	1.291	−3.843	0.545	0.281
exact	1.1745	−3.650	0.587	0.1745
2hom, one pair domain	1.153	−3.679	0.664	0.138

He	E
2split (two hemispheres)[a]	−3.026
exact	−2.9037
2hom (one pair, 1D radial)[b]	−2.845

[a] Axisymmetric; the pair repulsion is 4% below exact. [b] The Atoms paper [2]. (For H_2 the exact T is $-E$ by the virial theorem, and V_{eK} follows.) In both cases the exact value lies strictly between the two constructions, with no adjustable quantity on either side: $D_{2\text{hom}} < D_e < D_{2\text{split}}$ and $E_{2\text{split}} < E < E_{2\text{hom}}$. This is the model's energy statement. An intermediate construction that reproduced the exact value would be a fit.

The two constructions are distinct cases with no bridge between them. A pair either has a wall through it or it does not, and no parameter of the model carries one into the other. Charging the wall (Section 5.3) does not do it: as β grows the density moves away from the wall, not towards the shared pair. The situation is like a bifurcation. The pair sits on one branch or the other, and the states in between are not solutions of the model. The exact value lies between the two branches in energy, but it is not on a path that joins them.

The geometry is sharper than the energy. The 2split H_2 converges in the axisymmetric reduction to $R_e = 1.368\,a_0$, short of the exact 1.401 by 2.3% and flat to 0.2% under a 2.5-fold mesh refinement.

What makes the H_2 bond is not agreed among experts, even though the exact energy is not in doubt. One account, from Slater [7] and Feynman [8], is electrostatic: charge gathers between the nuclei, and the Hellmann–Feynman force on each nucleus pulls it inward. The other, from Hellmann [9] and Ruedenberg [10], puts the cause in the kinetic energy: an electron spread over both atoms lowers its kinetic energy, and the potential energy follows. The two accounts are still argued today [11]. Both read the same virial balance, $\Delta T = -\Delta E$ and $\Delta V = 2\Delta E$ at equilibrium, which holds whichever is the cause. The model takes no side by assumption. Its 2split bond is made by charge held in the bond region at no kinetic cost (Section 5.3), so it is closer to the electrostatic account. That is a property of the construction, not a settlement of the question.

That the cause of the simplest chemical bond is still argued, about ninety years after its energy was first computed, is not a failure of computation. The exact energy is not a fit: no coefficient is adjusted to experiment, and it corrected the measured dissociation energy rather than following it [13, 14]. Standard quantum mechanics explains the bond by law: the energy follows from the Schrödinger equation. It does not explain it by mechanism. The solution is a function on six-dimensional configuration space; a division of its energy into kinetic and potential causes depends on the reference state and the path chosen; and what happens in three dimensions is read into it, differently by different readers. Schrödinger asked for the second kind of explanation. A model in three dimensions carries a mechanism directly: the 2split bond is charge held in the bond region at a free wall that costs no kinetic energy. Its energy is not exact either, so the model does not settle the dispute; it shows what an explanation that can be checked looks like.

5.3 What places the 2split side

The 2split energy is not an error in the solution: two independent solvers agree, the energy converges, and every term is counted. Its components show what the free wall does. Against exact, the nuclear attraction is larger (centroid of electron 1 at −0.785, the nucleus at −0.70) and V_{ee}

smaller: a Neumann wall lets each electron keep full amplitude right up to the midplane—in the bond region, where both nuclei attract it—at no kinetic cost, and the other electron, confined strictly to the other side, never competes for that region. Charging the wall shows this directly. With a surface energy $\frac{\beta}{2}\int_{\partial\Omega_k} u_k^2$ on the interface:

β	T	V_{eK}	V_{ee}	centroid
0	1.291	−3.843	0.545	−0.785
0.15	1.277	−3.803	0.526	−0.832
0.5	1.265	−3.695	0.485	−0.936
2	1.281	−3.255	0.381	−1.269
10	1.238	−2.607	0.288	−1.713

As soon as the wall costs anything the density leaves the interface for the far side of its own nucleus and the nuclear attraction falls, with the kinetic energy almost unchanged. The free cut at the wall is the model's central assumption; it sets the 2split energy, while the 2hom pair has no wall.

5.4 Two states of a pair

The two constructions are two different states of the pair, and standard quantum mechanics has both. The 2hom pair, two electrons of opposite spin carrying one density over one domain, is the restricted Hartree–Fock singlet: one doubly occupied orbital. In H_2 at $R = 1.4$ the axisymmetric 2hom energy is -1.133585 at $h = 0.1$, against the Hartree–Fock limit -1.133630 [12]. The two share the known Hartree–Fock behaviour: H^- is unbound, and the pair does not dissociate into two neutral atoms, because half of it stays ionic. The 2split pair is the Heitler–London state. Each electron keeps to its own region, which is avoidance in the extreme, and spin plays no part: without an exchange term the singlet and the triplet are the same. It dissociates correctly and binds H^-. The exact ground state is neither. It is a correlated singlet, which standard quantum mechanics builds by mixing the shared and the separated configurations. A real density with one construction per pair has no such mixture, and that is why the exact energy lies between the two and is reached by neither (Section 5).

In Hartree–Fock the two states are joined by a bifurcation. The restricted solution is stable near equilibrium and becomes unstable beyond the Coulson–Fischer point, about $R = 2.3$, where it breaks symmetry into the unrestricted solution, which separates the electrons like 2split [15]. The model has the two branches but not the path between them. Domains are disjoint, and two electrons share a domain only with the same density, so the partial overlap through which the Hartree–Fock solution breaks symmetry is not available. A 2hom pair therefore cannot decay into a 2split pair. It is also lower in energy at no R:

R	D(2hom)	D(2split)	$E_{\text{split}} - E_{\text{hom}}$
1.0	0.0847	0.2305	−0.1458
1.2	0.1246	0.2679	−0.1434
1.4	0.1332	0.2747	−0.1416
1.6	0.1259	0.2663	−0.1404
2.0	0.0912	0.2308	−0.1396
2.4	0.0489	0.1894	−0.1406
3.0	−0.0111	0.1336	−0.1447
4.0	−0.0887	0.0679	−0.1566
5.0	−0.1410	0.0308	−0.1718
6.0	−0.1755	0.0114	−0.1869
8.0	−0.2135	−0.0037	−0.2098

(Axisymmetric, $h = 0.1$, $D = 2E_{\mathrm{H}} - E$ on the same mesh; 2hom mirror-symmetric.) The two curves do not cross: 2split is lower at every R, by 0.14 near equilibrium and by more as the bond stretches. If a path existed, energy minimisation would take every pair to 2split, including a bonding pair at equilibrium, where 2hom is the closer one to experiment (Section 5.5). The free wall over-favours avoidance. So the construction of a pair is not chosen by energy. It is chosen by the physical rule of the next subsection: shared pairs are 2hom, and electrons that must avoid each other are 2split.

Breaking a bond requires 2split. The scan shows the division of labour directly. The 2hom pair stops binding near $R = 3$ and rises steeply beyond it, 0.21 above two atoms at $R = 8$, because half of the shared pair stays ionic, as in restricted Hartree–Fock. Left without the mirror symmetry, a stretched 2hom pair does not turn into a 2split pair. It collapses onto one nucleus, the ionic state $\mathrm{H}^- + \mathrm{H}^+$ (97.5% of the density on one side at $R = 6$), which is a higher solution than the symmetric pair. Only 2split goes to two neutral atoms. Its tail falls off too slowly, however: it still binds 0.011 at $R = 6$, where the exact binding is nearly zero, and it is slightly repulsive (0.004) at $R = 8$. This matters for the later aim of reactions between fragments. A reaction breaks or forms bonds, so somewhere along its path a pair has to change from 2hom to 2split, and the model has no path between them. The energies at the two ends of a reaction can be computed in the model; a barrier between them needs a rule for where the pair changes, which the model does not yet have.

A bridge by competition. Reaction–diffusion systems suggest a bridge. Two densities that diffuse and compete through a term $\kappa u_1^2 u_2^2$ separate into disjoint regions as $\kappa \to \infty$, the strong-competition limit [16]. We give each electron its own density over the whole space and add $\kappa \int u_1^2 u_2^2$ to the energy. At $\kappa = 0$ this is unrestricted Hartree–Fock for two opposite spins: the Hartree repulsion between two orbitals, with no exchange term. Started each time from the electrons on opposite sides (axisymmetric, $h = 0.1$):

R	κ	D	orbital overlap	state
1.4	0	0.1332	1.000	symmetric = 2hom
1.4	5	−0.0429	1.000	symmetric
1.4	6	−0.0707	0.973	broken
1.4	1000	−0.5604	0.045	segregated
2.2	0	0.0702	1.000	symmetric
2.4	0	0.0497	0.923	broken
4.0	0	0.0012	0.265	broken

The family has a pitchfork bifurcation. At $\kappa = 0$ the symmetric solution breaks between $R = 2.2$ and 2.4, which is the Coulson–Fischer point. At $R = 1.4$ it breaks between $\kappa = 5$ and 6. Near both breaks the iteration slows down, as it does at a bifurcation. In the (R, κ) plane a curve of symmetry breaking separates the shared pair from separated electrons, and the broken $\kappa = 0$ branch dissociates correctly ($D = 0.001$ at $R = 4$). The far end of the family, however, is not 2split. Segregation by competition forces the densities to vanish at the interface, and this hard wall costs energy: at $R = 1.4$, $\kappa = 1000$ is 0.56 above two atoms, while 2split binds by 0.275. No overlap penalty reaches the free Neumann wall. The wall is the model's own assumption and the source of its over-binding (Section 5.3), and 2split stands outside the family of overlapping states. A switching rule by competition is therefore available, but it switches into the hard-wall branch. Using it for reactions would add these states to the model rather than join its two constructions.

5.5 Beyond two electrons

The same two constructions apply to any pair in a molecule. For LiH, with Li as a +1 pseudo-nucleus ($r_c = 1.99$ from its ionisation energy) and the bonding pair as 2split (Li valence and H electron in separate domains) or 2hom (one pair over Li^+ and a bare proton, Section 6.1), D_e is 0.063–0.073 (2split) and 0.095 (2hom) against the exact 0.092. For BH (pseudo-B +3 with a 2hom lone pair) D_e is 0.276 (2split) and 0.155 (2hom) against 0.135 (Table 1). In both, the 2hom bonding pair is the closer.

The picture also tells where each construction belongs. A pair that shares one region—a core $1s^2$, or a bonding pair near equilibrium—is naturally 2hom. Electrons that must avoid each other need 2split: the 2hom H^- is unbound, exactly as in Hartree–Fock, while the 2split H^- binds; and a 2hom pair cannot dissociate a bond into two neutral atoms. We report both, and choose by this rule.

The results below divide the roles as follows. *Geometry is common to both cases:* where both were run, the bond lengths differ by a few per cent at most (Li_2 5.1/5.15, N_2 1.95/1.95, LiH 2.85/2.98, BH 2.29/2.38; Table 1 and Section 6.3), so the bond length does not depend on the choice. *2split is the general case and over-binds:* its D_e is above experiment in most molecules computed, up to about twice (BH 0.276 against 0.135, OH 0.371 against 0.170). The exceptions are LiH (0.063–0.073 against 0.092) and B_2 (at experiment). *2hom is the shared pair:* two opposite spins in one domain. It is the closer energy for a shared pair at equilibrium (LiH 0.095 against 0.092, C_2 0.242 against 0.232, N_2 0.370 against 0.364) and under-binds where the electrons should avoid each other. It also gives the LiH dipole, of the right sign and 11% high, where one electron per domain inverts it. Dipoles in general are not settled by either case. In BH only a cone-shaped boundary corrects the dipole, at the cost of bond length and binding (Section 6.2). In OH both cases are about twice too large. H_2O needs split O for the right sign.

6 Results

The molecules below are a first selection of basic examples, chosen where the model's constructions are settled; many more are to follow, and those tried beyond the selection are collected in Section 7. The criterion is geometry: bond lengths. Binding energies are listed as a diagnostic of the construction, not as a target. **[In progress.]** Bond lengths from an R-scan at three meshes ($h = 0.10, 0.07, 0.05$) per bond with `molecule.js`, extrapolated in h, with the kernel clamp pinned so that the ladder refines a fixed problem; axisymmetric values beside them for X_2 only. Energies for both constructions, 2hom and 2split, at the computed geometry.

		bond length (a_0)				binding D_e (E_h), diagnostic		
molecule	construction	3D	axisym	exp.	error	2hom	2split	exp.
H_2	1+1	pending	1.368	1.401	−2.3%	0.138	0.281	0.1745
LiH	Li^+ (r_c) + pair	2.85	—	3.015	−5%	0.095	0.063–0.073	0.092
BH	B 2+3 (2_{hom}+1)	2.29	—	2.329	−1.6%	0.155	0.276	0.135
OH	O +3 (2_{hom}+1)	1.79	—	1.832	−2%	0.241	0.371	0.170
BeH_2	Be $2+2_{split}$	2.44	—	2.52[a]	−3%	0.248[b]	n.c.	0.2231

Table 1: Bond lengths and the 2hom and 2split binding energies against experiment. H_2 energies are axisymmetric ($R = 1.4$, $h = 0.1$); LiH is three-dimensional (box $20\,a_0$, $h = 0.1$), its bond length from the 2hom construction (2split: 2.98–2.99). BH is three-dimensional (box $12\,a_0$, $h = 0.1$, pseudo-B $r_c = 1.08$), $R = 2.1, 2.3, 2.5$; bond length from 2hom (2split: 2.38). BeH_2: pseudo-Be $r_c = 1.37$, box $12\,a_0$, $R = 2.3, 2.5, 2.7$; [a]gas-phase r_0 (vibrationally averaged); [b]atomisation energy (two bonds), experiment $0.2231\,E_h$ [6]; n.c.: the 2split runs do not converge (the two sides trade cells and the symmetry breaks). OH: pseudo-O +3 (only the outer 3-shell carried, as 2hom lone pair +1 behind a plane), $r_c = 0.59$, $R = 1.65, 1.83, 2.0$; 2split $R_e \approx 1.69$; dipole 3.6 D (2hom), 2.7 D (2split) against 1.66. The axisymmetric reduction is used for X_2 and for linear molecules with fixed walls only.

6.1 LiH: pseudo-Li and one bonding pair

LiH is the simplest polar bond with an inner shell, and is viewed as ionic (Li^+H^-) rather than covalent: its dipole, 5.88 D, is 77% of that of point charges ±1 at $3.015\,a_0$ (7.66 D). The model reduction reflects this: the Li $1s^2$ core is excluded by a fixed sphere of radius $r_c = 1.99\,a_0$ around the Li nucleus, which then carries charge +1 (pseudo-Li); the two valence electrons form one 2hom pair over pseudo-Li and a bare proton. The radius is the one at which the pseudo-Li atom alone reproduces the Li ionisation energy; the one-dimensional all-electron Li atom places its own core–valence interface at 1.98–$2.05\,a_0$ over boxes 6–$20\,a_0$, within 3% of it. All runs are `molecule.js`, $h = 0.1$, box $20\,a_0$; binding energies are $E(\mathrm{Li}) + E(\mathrm{H}) - E(\mathrm{LiH})$ with both atoms built the same way and in the same box.

R	E	$D(R)$	μ (D)
2.6	−0.76776	0.0880	—
2.8	−0.77436	0.0946	6.45
3.0	−0.77259	0.0928	6.53
3.2	−0.76709	0.0873	6.66
3.4	−0.75999	0.0802	6.82
exp.		0.092 at 3.015	5.88

A fit through the five points gives $R_e = 2.85\,a_0$ (2.81–2.89 over three-, four- and five-point fits), 5% short of experiment, and $D_e = 0.0947\,E_h$, 3% over. At $R = 3.0$ the dipole is 6.53 D, of the right sign (Li^+H^-) and 11% high; it is 86% of the point-charge value at that R, so the pair, free to distribute itself over both centres, settles largely on H. Nothing was adjusted to LiH. The 2split on the same pseudo-Li (Li valence and H electron in separate domains) gives a slightly longer bond, $R_e = 2.98$–2.99 (−1%), and under-binds, $D_e = 0.063$–0.073 (Section 5.5). The core is a hard exclusion with a radius set by the atom: a frozen-core model, stated as such. A 2hom core held inside the same fixed sphere is equivalent for the valence, since a spherical charge inside the sphere acts outside it as a point charge. The sphere must be fixed: a free boundary between core and

valence charges nothing for crowding, and with it the valence presses into the core and $E(R)$ falls monotonically as H approaches.

6.2 BH and BeH_2

BH. Pseudo-B +3 carries its outer 2+3 shell as a 2hom lone pair and one electron, the two separated by a fixed surface through the B nucleus; the B|H boundary is free. Two divisions of the 3 were run, each with its own r_c fitted to the B ionisation energies (box $12\,a_0$, $h = 0.1$): a plane perpendicular to the bond (each domain half the solid angle, $r_c = 1.08$), and a cone about the bond holding one third of the solid angle for the single electron and two thirds for the pair (half-angle 70.5°, $r_c = 1.11$), in proportion to their electrons. For the bond, 2hom (one pair over the B front and a bare proton):

2hom BH	R_e	D_e	μ (D)
plane, $\frac{1}{2}$:$\frac{1}{2}$	2.29	0.155	3.5
cone, $\frac{1}{3}$:$\frac{2}{3}$	2.17	0.224	1.09
exp.	2.329	0.135	1.27

The proportional division corrects the dipole, by moving charge behind B, and costs bond length and binding: the pair confined to the cone draws the proton in. The plane gives bond and energy, the cone the dipole; neither gives all three. 2split binds more strongly in both (0.276 and about 0.34), with dipoles 4.0 and 1.2–1.3 D. Table 1 uses the plane.

BeH_2. Pseudo-Be +2 (2+2, $r_c = 1.37$) carries its two valence electrons as single electrons on either side of a fixed plane through Be, each facing one H. With each bond a 2hom pair over a Be half and a bare proton, $R_e = 2.44$ against the gas-phase $r_0 = 2.52$ (−3%), and the atomisation energy is $0.248\,E_h$ against 0.2231 [6] (+11%). The 2split construction does not settle: the two H domains and the two Be halves trade cells without end, the symmetry breaks (a residual dipole of 0.2 D), and the convergence window never closes.

6.3 Homonuclear X_2

For X_2 the axisymmetric solver is used. Each X is a pseudo-atom whose inner shells, as given by the Atom Simulator configuration, are frozen into a core of charge Z_x with r_c fitted to the sum of the ionisation energies of the electrons carried; the outer shell lies along the axis as a domain behind X (a 2hom lone pair, or one electron) and one bonding electron, separated by a fixed plane through X. The two bonding electrons form either a 2split pair (the midplane) or one 2hom domain between the nuclei. Only a σ pair is representable. R-scan 0.8–1.3 R_{exp}, $h \approx 0.1$ (adjusted per R to keep the nuclei on cell faces, a jitter of about $\pm 0.005\,E_h$):

X_2	configuration	r_c	R_e: 2hom	/ 2split	D_e: 2hom	/ 2split	exp. R_e / D_e
Li_2	2+1, +1	2.00	5.1	5.15	0.021	0.054	5.05 / 0.038
Na_2	2+4+4+1, +1	2.16	5.5	5.55	0.020	0.050	5.82 / 0.028
B_2	2+2+1, +1	0.88	2.87	2.9	0.045	0.109	3.005 / 0.110
C_2	2+2+2, +2	0.91	2.55	2.58	0.242	0.322	2.348 / 0.232
N_2	2+3+2, +2	0.59	1.95	1.95	0.370	0.480	2.074 / 0.364
O_2	2+3+3, +3	0.59	2.0	2.05	0.349	0.48	2.282 / 0.192
F_2	2+4+3, +3	0.49	no minimum		> 0.35 / > 0.52		2.668 / 0.061

For Li_2, Na_2, B_2, C_2 and N_2, cores +1 and +2, the experimental binding lies between the two constructions and the bond length is within 10%; B_2 and N_2 require the Atom Simulator configurations (as pseudo-+3 with a lone pair, B_2 over-binds twofold). O_2 and F_2 are not reproduced in any configuration tried (Section 7).

Transfer of the carbon model: C_2H_2. The C_2 carbon (pseudo-C +2, $r_c = 0.91$, valence 1+1) carried unchanged into acetylene, with every bond a 2hom pair (each C–H pair the H electron and the outer C half over a bare proton; the C–C pair the two facing halves), has all its walls fixed at the planes through the C nuclei, so the axisymmetric solution is that of the three-dimensional construction. Each bond scanned with the other at its observed length:

C_2H_2	C–C	C–H	atomisation at obs. geometry
2hom	2.50 (+10%)	2.0 (0%)	0.573–0.579
2split*	2.55 (+12%)	1.95 (−3%)	0.896
exp.	2.274	2.009	≈ 0.646

*The 2split C|H walls are free and are held plane, an approximation. The experimental atomisation lies between the two constructions, 2hom the closer (−11%); the C–H bond is right, and the C–C bond is 10% long, as in C_2 (+9%). The carbon model transfers with a consistent C–C bias, plausibly the single σ pair standing in for multiple bonding. Carrying the carbon valence instead as one 2hom pair per atom ($r_c = 0.81$) gives C_2 the right length (+2%) but $D_e = 0.359$ (+55%), a binding of two closed pairs across a free wall that charges nothing, and is not used.

Other linear molecules. The same pseudo-atoms, unchanged (C +2, N +2 with one electron behind, O +3 with a 2hom lone pair behind; each bond one 2hom σ pair, a fixed plane at every heavy nucleus), give the following linear molecules (axisymmetric, $h \approx 0.1$; R_e from a parabola through the scan about R_{exp}; atomisation at the observed geometry):

molecule	bond	R_e (exp.)	atomisation (exp.)
HCN	C–N	2.25 (2.179), +3%	0.475 (≈ 0.50)
	C–H	1.97 (2.014), −2%	
CO	C–O	2.19 (2.132), +3%	0.344 (0.413)
CO_2	C–O, both	2.15 (2.192), −2%	0.681 (0.620)
CH	C–H	1.93 (2.116), −9%	0.181 (≈ 0.134)
CH_2, linear	C–H, both	1.97 (2.03, bent)	0.355 (≈ 0.30)

Where carbon sits between two partners on the axis the bonds are within 3%, even with the +3 oxygen of CO and CO_2, whose lone pairs face away from each other, unlike those of O_2. Where carbon ends the chain with an electron behind it (CH, linear CH_2) the C–H bond is short: the remaining carbon electrons of the real radical lie off the axis, which a σ-only model cannot represent.

A chain released. The scans above hold every other bond at its observed length. Released as a whole from the observed geometry, cyanoacetylene HC≡C–C≡N relaxes in all four bonds together (coordinate descent at fixed $h = 0.05$, steps of $0.1\,a_0$, a parabola per bond):

bond	observed (r_s)	relaxed	change
C–H	1.997	1.993	−0.2%
C≡C	2.273	2.560	+12.6%
C–C	2.604	2.545	−2.3%
C≡N	2.190	2.162	−1.3%

The bonds whose type is set by the atoms (C–H, C≡N) stay within 1.3%; the two carbon–carbon bonds come out nearly equal. The chain construction gives every bond one 2hom pair whatever its order, with the rest of carbon's valence frozen into the +2 core, so a triple and a single C–C bond are the same construction and relax to the same length. Bond alternation, the difference between multiple and single bonds along a chain, is not represented. This is the same defect as the 10%-long C≡C of C_2H_2; the bonds of CO_2, all of one kind, are within 2%.

6.4 Forces at the observed geometry

The force on a nucleus is read as the pull of the converged density, $\mathbf{F}_a = Z_a \int \rho\,(\mathbf{x}-\mathbf{R}_a)/|\mathbf{x}-\mathbf{R}_a|^3$ outside the kernel clamp, plus the nuclear repulsion. It equals $-\partial E/\partial \mathbf{R}_a$ only if the energy is stationary in everything that moves with the nucleus. A bare proton inside a 2hom bond pair carries no wall of its own, and there the readout agrees with the slope of $E(R)$ within 0.002–0.011 (LiH, BeH_2, BH; also CH_4, Section 7). With 2split the H domain's wall moves with H and is placed by continuity, not by the energy, and the readout misses that wall term (BH: −0.040 against +0.086). Forces are therefore read with bond pairs and bare protons; the spread between equivalent H atoms, about ±0.005, is the mesh noise at $h = 0.1$.

At the observed geometry the residual force on each H, along the bond (positive: the bond would lengthen) is, in E_h/a_0:

molecule	geometry	along bond
LiH	3.0	−0.022
BH	2.3	−0.013
BeH_2	2.5	−0.018
OH	1.83	−0.028
axisymmetric, $-\mathrm{d}E/\mathrm{d}R$ over ±10% of the bond:		
HCN	C–N / C–H	+0.042 / −0.013
CO	C–O	+0.045
CO_2	C–O, per bond	−0.022
C_2H_2	C–C	+0.19
CH	C–H	−0.046

Every hydride has a small inward force at the observed geometry, 0.013–0.028: the model's X–H bonds are slightly short. A force must be read against the stiffness: the triple bond of C_2H_2 carries +0.19 for a 10% error, CO_2 −0.022 for 2%.

7 Scope, and cases that do not yet work

Scope. Inner shells cannot be carried as domains with free walls: the valence presses into them and the bond collapses (every all-electron LiH with free core walls does, Section 6.1). They are excluded by a fixed core radius per atom, set by the atom's ionisation energies; this is a frozen-core model. The axisymmetric reduction carries only pairs symmetric about the axis. Contacts between molecules add a new free wall in a region of high density, and dispersion has no counterpart; intermolecular energies are outside the present model.

Cases that do not yet work. The selection above is a first one. The following were tried beyond it; each is stated with the result found:

- *Heavy atoms carrying their whole valence* (CH_4, NH_3, H_2O): bonds a few per cent short (CH_4 -4%); binding two to three times too large, by an amount per heavy atom (ethane twice methane).
- *Shapes and angles:* the CH_4 tetrahedron and square plane have the same energy within $0.002\,E_h$ (true difference about 0.2); the H_2O angle closes below 95° with the lone pairs frozen into the core and is pushed open with them as domains of their own.
- *O_2 and F_2:* O_2 over-binds about twofold with a bond 12% short; F_2 has no minimum; no core configuration or radius tried reproduces either.
- *Bond order:* each bond is one pair, so the triple and single C–C bonds of HC≡C–C≡N relax to nearly equal lengths (Section 6.3).
- *Dipoles:* LiH right; BH, OH and H_2O off in sign or size; NaH and HF not yet run.
- *Large molecules:* within computational reach, not yet validated (Section 3).
- *Reaction energies:* quotable only where every atom keeps its bonding; an isodesmic test among linear molecules alternates by about ± 2.5 kcal/mol between meshes, below what is needed.

Most of these share one feature: where pairs are pressed together the free wall between them charges nothing, and the model has no repulsion between pairs beyond their Coulomb energy. Whether RealQM needs such a repulsion, and in what form, is the main open question; its test is F_2 weakly bound near $2.67\,a_0$, the H_2O angle and the CH_4 tetrahedron recovered, and LiH, Li_2, C_2, N_2 left where they are.

8 Conclusion

first selection	status
H_2, He	exact value between 2split and 2hom; H_2 bond -2.3%
hydrides LiH, BH, BeH_2, OH	bonds 2–5% short; residual forces inward, 0.013–$0.028\,E_h/a_0$; binding within 0.003–$0.3\,E_h$; LiH dipole right sign
X_2	Li_2, Na_2, B_2, C_2, N_2: bond within 10%, binding between the two constructions
linear molecules	HCN, CO, CO_2 within 3%; C_2H_2 C≡C 10% long
beyond the selection	open questions with results so far (Section 7): heavy atoms with their whole valence, shapes and angles, O_2 and F_2, bond order, dipoles, reaction energies

Solved as it stands, the hard-partition model of RealQM carries every electron pair in one of two parameter-free constructions, 2split and 2hom, the limits of complete and of no avoidance; for H_2 and He the exact energy lies between them. A shared pair (core, bonding pair) is 2hom, and for LiH and BH the 2hom bond is the closer to experiment. Inner shells must be excluded by a fixed core radius, since a free wall charges nothing for crowding; with that, LiH as one 2hom bonding pair has its bond within 5%, its binding within 3% and a dipole of the right sign. Nothing is adjusted to molecular data; each atom contributes one core radius, set from its ionisation energies.

On this first selection of basic examples the model is a geometry model: bond lengths of hydrides, of X_2 from Li_2 to N_2 and of linear molecules built from the same pseudo-atoms are within a few

per cent, with small residual forces at the observed geometry. Binding energies are a diagnostic. Molecules beyond the selection are not yet reproduced or only with low accuracy, and are left as open questions with the results obtained so far; most point to one of them, whether a repulsion between pairs pressed together is needed. Energies above five domains rest on an independent recomputation from the converged density, which is how a fault in the solver's default electron–electron path was found. The validation continues: the selection will be extended molecule by molecule, and the cases that do not yet work are the next steps.

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